MOD 1

(Pages : 2)

Name.....

Reg. No.....

FYUGP B.Sc. PHYSICS HONOURS

ELECTRODYNAMICS II

Semester V · Core in Major

MODULE 1: ELECTRIC AND MAGNETIC FIELDS IN MATTER

Past questions

Section A: 1 to 3 marks

Polarization and dielectrics

1 Is there any difference between dielectric constant and dielectric strength ? **(1 mark)** *April 2018*

2 State whether true or false: CO_2 is an example of non polar molecule. **(1 mark)** *April 2020*

3 Is dielectric an insulator ? Explain. **(2 marks)** *April 2020*

4 What is meant by dielectric breakdown ? **(2 marks)** *April 2020*

5 Write down the relation connecting polarization and electric field in linear dielectrics. What do you mean by a polar dielectric ? **(3 marks)** *November 2021*

6 Give the relation connecting permittivity and susceptibility in linear dielectrics. Explain the terms involved. How is relative permittivity connected to susceptibility ? **(3 marks)** *November 2021*

7 Write down the expression for the torque experienced by a dipole in uniform electric field. What is the effect of the torque ? **(2 marks)** *November 2022*

8 Write down the relation connecting permittivity and susceptibility in linear dielectrics. Explain the terms involved. What is the susceptibility of vacuum ? **(2 marks)** *November 2022*

Bound charge, D and dielectric capacitors

9 For a given potential difference does a capacitor store more or less charge with a dielectric than it does without a dielectric ? **(1 mark)** *April 2018*

10 What is bound charge ? **(1 mark)** *April 2018*

11 State whether true or false: In the interior of a dielectric kept in a uniform field, the net charge is zero. **(1 mark)** *April 2020*

12 Write down Gauss's law in presence of a dielectric. **(3 marks)** *November 2021*

Magnetization and magnetic materials

13 State whether true or false: Above the Curie temperature, a ferromagnetic material would become paramagnetic. **(1 mark)** *April 2020*

14 Distinguish between permeability and permittivity. **(2 marks)** *April 2020*

15 What is the basic difference between electric and magnetic polarizations ? **(3 marks)** *November 2021*

16 When will you say that a medium is magnetically polarized ? Discuss the magnetic polarization in diamagnets. **(2 marks)** *November 2022*

17 Explain the temperature-driven phase transitions in ferromagnetic materials. **(2 marks)** *November 2022*

Bound currents, H and Ampère in matter

18 What is bound currents ? **(2 marks)** *April 2018*

19 Discuss magnetic field inside matter. **(2 marks)** *April 2018*

20 Write the relation connecting $\boldsymbol{M}$, $\boldsymbol{B}$ and $\boldsymbol{H}$. **(2 marks)** *April 2020*

Section B: 4 to 5 marks

Polarization and dielectrics

21 Obtain Clausius–Mossotti formula. **(4 marks)** *April 2018*

22 Dielectric constant of a gas at NTP is 1.000074. Calculate dipole moment of each atom of the gas when it is held in an external field of $3 \times 10^4\ \mathrm{V\,m^{-1}}$. **(4 marks)** *April 2018*

23 Show that the energy of an ideal dipole $\vec{p}$ in an electric field $\vec{E}$ is given by $U = -\vec{p} \cdot \vec{E}$. **(5 marks)** *November 2021*

Bound charge, D and dielectric capacitors

24 Derive the expression for capacitance of a parallel plate capacitor with partially filled dielectric. **(4 marks)** *April 2018*

25 Compare the curl of $\vec{E}$ and $\vec{D}$ in electrostatics. **(5 marks)** *November 2022*

Boundary conditions

26 Are the electric field and magnetic field continuous at a boundary between two different media ? Find the relevant equations. **(4 marks)** *November 2019*

27 Explain electrostatic boundary conditions. **(4 marks)** *April 2020*

28 Write down the boundary conditions for the electric and magnetic vectors $\vec{E}$ and $\vec{B}$ at an interface separating two linear dielectrics of permittivities ε_1 and ε_2 and permeabilities μ_1 and μ_2. **(4 marks)** *November 2021*

Magnetization and magnetic materials

29 Derive an expression for the gyro magnetic ratio. **(4 marks)** *April 2018*

30 Explain the domain model of ferromagnets. **(5 marks)** *November 2021*

31 Explain the hysteresis loop of ferromagnets. **(5 marks)** *November 2022*

Bound currents, H and Ampère in matter

32 In the Rowland ring 2.0 amp current is passing through the winding of number of turns per unit length 10 turns/cm. Magnetic induction measured is $1.0\ \mathrm{weber/m^2}$. Calculate (a) H (b) M. **(4 marks)** *April 2018*

33 A toroidal winding of N turns surrounding a ferromagnetic specimen in which a narrow gap of width d has been cut. Calculate the value of magnetic field both in the gap and in the material. **(4 marks)** *April 2018*

34 Write an expression for the field of a magnetized object. Explain the terms involved. **(5 marks)** *November 2021*

35 Give an expression for the field of a magnetized object. Explain the physical meaning of the expression. **(5 marks)** *November 2022*

Section C: 10 marks

Bound charge, D and dielectric capacitors

36 Explain the terms polarization and displacement. Derive the relation connecting them. Derive an expression for potential of polarized objects. **(10 marks)** *April 2018*

37 Show that the potential due to a polarized dielectric is the same as that produced by a volume charge density and surface charge density. **(10 marks)** *April 2020*

Magnetization and magnetic materials

38 Explain the following types of magnetic materials: (a) Paramagnets; (b) Diamagnets; and (c) Ferromagnets. **(10 marks)** *April 2020*

Bound currents, H and Ampère in matter

39 Discuss in detail the Ampere's law in Magnetized materials. Derive an expression for the magnetic field at any point inside and outside of a long copper rod of radius R carries a uniformly distributed current I. **(10 marks)** *April 2018*